Statistical Modelling

Exam Number: B261807

2026-06-27

Part I

Carry out a simple exploratory investigation of baseline variables (using appropriate numerical summaries and graphical methods). You should explain the implications of your findings with respect to causal concepts and downstream modelling assumptions.

```
## # A tibble: 10 x 17
##    patient_id   age sex    smoking_status deprivation_quintile comorbidity_count
##         <dbl> <dbl> <chr>  <chr>          <chr>                            <dbl>
##  1          1  54   Male   Never          Q4                                   0
##  2          2  47.9 Male   Current        Q3                                   1
##  3          3  79   Female Current        Q3                                   0
##  4          4  49.9 Female Never          Q5                                   0
##  5          5  81.7 Female Never          Q1                                   3
##  6          6  75.6 Female Never          Q4                                   0
##  7          7  46.6 Female Never          Q5                                   1
##  8          8  71.1 Male   Former         Q1                                   2
##  9          9  58.9 Male   Never          Q1                                   1
## 10         10  52.3 Female Former         Q3                                   2
## # i 11 more variables: prior_exacerbations_12m <dbl>,
## #   baseline_symptom_score <dbl>, baseline_fev1_pct <dbl>, new_therapy <chr>,
## #   followup_months <dbl>, symptom_change_6m <dbl>,
## #   followup_symptom_score <dbl>, severe_exacerbation_12m <dbl>,
## #   gp_exacerbations_12m <dbl>, time_to_event_days <dbl>, followup_event <dbl>
```

```
## spc_tbl_ [620 x 17] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ patient_id             : num [1:620] 1 2 3 4 5 6 7 8 9 10 ...
##  $ age                    : num [1:620] 54 47.9 79 49.9 81.7 75.6 46.6 71.1 58.9 52.3 ...
##  $ sex                    : chr [1:620] "Male" "Male" "Female" "Female" ...
##  $ smoking_status         : chr [1:620] "Never" "Current" "Current" "Never" ...
##  $ deprivation_quintile   : chr [1:620] "Q4" "Q3" "Q3" "Q5" ...
##  $ comorbidity_count      : num [1:620] 0 1 0 0 3 0 1 2 1 2 ...
##  $ prior_exacerbations_12m: num [1:620] 1 2 1 0 2 0 3 1 0 1 ...
##  $ baseline_symptom_score : num [1:620] 22 24.1 24.2 14.1 26.4 20 20.8 21.6 22 30.7 ...
##  $ baseline_fev1_pct      : num [1:620] 58.3 79.9 52 62.6 67 70.2 77.5 61.1 60.4 56.9 ...
##  $ new_therapy            : chr [1:620] "Usual care" "Usual care" "New therapy" "New therapy" ...
##  $ followup_months        : num [1:620] 10.1 14.6 13.2 11 9.6 15 14.1 13.8 11.8 12.2 ...
##  $ symptom_change_6m      : num [1:620] 4.5 1.8 11.1 5.9 -2.3 -0.9 -1.3 -0.7 -8.2 -5.8 ...
##  $ followup_symptom_score : num [1:620] 17.5 22.3 13.1 8.2 28.7 20.9 22.1 22.3 30.2 36.5 ...
##  $ severe_exacerbation_12m: num [1:620] 0 1 0 0 1 0 1 0 0 0 ...
##  $ gp_exacerbations_12m   : num [1:620] 2 6 1 2 3 1 5 1 1 3 ...
```

```
##  $ time_to_event_days     : num [1:620] 275 272 317 279 505 571 218 237 167 623 ...
##  $ followup_event         : num [1:620] 1 1 1 1 1 1 1 1 1 1 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   patient_id = col_double(),
##   ..   age = col_double(),
##   ..   sex = col_character(),
##   ..   smoking_status = col_character(),
##   ..   deprivation_quintile = col_character(),
##   ..   comorbidity_count = col_double(),
##   ..   prior_exacerbations_12m = col_double(),
##   ..   baseline_symptom_score = col_double(),
##   ..   baseline_fev1_pct = col_double(),
##   ..   new_therapy = col_character(),
##   ..   followup_months = col_double(),
##   ..   symptom_change_6m = col_double(),
##   ..   followup_symptom_score = col_double(),
##   ..   severe_exacerbation_12m = col_double(),
##   ..   gp_exacerbations_12m = col_double(),
##   ..   time_to_event_days = col_double(),
##   ..   followup_event = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```
## Rows: 620
## Columns: 17
## $ patient_id              <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,~
## $ age                     <dbl> 54.0, 47.9, 79.0, 49.9, 81.7, 75.6, 46.6, 71.1~
## $ sex                     <chr> "Male", "Male", "Female", "Female", "Female", ~
## $ smoking_status          <chr> "Never", "Current", "Current", "Never", "Never~
## $ deprivation_quintile    <chr> "Q4", "Q3", "Q3", "Q5", "Q1", "Q4", "Q5", "Q1"~
## $ comorbidity_count       <dbl> 0, 1, 0, 0, 3, 0, 1, 2, 1, 2, 3, 1, 2, 0, 0, 1~
## $ prior_exacerbations_12m <dbl> 1, 2, 1, 0, 2, 0, 3, 1, 0, 1, 3, 0, 0, 1, 0, 1~
## $ baseline_symptom_score  <dbl> 22.0, 24.1, 24.2, 14.1, 26.4, 20.0, 20.8, 21.6~
## $ baseline_fev1_pct       <dbl> 58.3, 79.9, 52.0, 62.6, 67.0, 70.2, 77.5, 61.1~
## $ new_therapy             <chr> "Usual care", "Usual care", "New therapy", "Ne~
## $ followup_months         <dbl> 10.1, 14.6, 13.2, 11.0, 9.6, 15.0, 14.1, 13.8,~
## $ symptom_change_6m       <dbl> 4.5, 1.8, 11.1, 5.9, -2.3, -0.9, -1.3, -0.7, -~
## $ followup_symptom_score  <dbl> 17.5, 22.3, 13.1, 8.2, 28.7, 20.9, 22.1, 22.3,~
## $ severe_exacerbation_12m <dbl> 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1~
## $ gp_exacerbations_12m    <dbl> 2, 6, 1, 2, 3, 1, 5, 1, 1, 3, 7, 2, 2, 4, 3, 2~
## $ time_to_event_days      <dbl> 275, 272, 317, 279, 505, 571, 218, 237, 167, 6~
## $ followup_event          <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1~
```

```
##    patient_id         age            sex            smoking_status
##  Min.   :  1.0   Min.   :19.00   Length:620         Length:620
##  1st Qu.:155.8   1st Qu.:48.98   Class :character   Class :character
##  Median :310.5   Median :58.05   Mode  :character   Mode  :character
##  Mean   :310.5   Mean   :58.29
##  3rd Qu.:465.2   3rd Qu.:67.20
##  Max.   :620.0   Max.   :88.00
##  deprivation_quintile comorbidity_count prior_exacerbations_12m
##  Length:620           Min.   :0.000     Min.   :0.000
##  Class :character     1st Qu.:0.000     1st Qu.:0.000
```

```
##  Mode  :character     Median :1.000     Median :1.000
##                       Mean   :1.216     Mean   :1.258
##                       3rd Qu.:2.000     3rd Qu.:2.000
##                       Max.   :5.000     Max.   :7.000
##  baseline_symptom_score baseline_fev1_pct new_therapy        followup_months
##  Min.   : 4.30          Min.   : 25.00    Length:620         Min.   : 9.00
##  1st Qu.:20.70          1st Qu.: 55.67    Class :character   1st Qu.:10.60
##  Median :24.90          Median : 63.05    Mode  :character   Median :12.20
##  Mean   :24.89          Mean   : 62.80                       Mean   :12.14
##  3rd Qu.:29.23          3rd Qu.: 70.12                       3rd Qu.:13.70
##  Max.   :40.00          Max.   :102.60                       Max.   :15.00
##  symptom_change_6m  followup_symptom_score severe_exacerbation_12m
##  Min.   :-16.3000   Min.   : 2.90          Min.   :0.0000
##  1st Qu.: -2.4000   1st Qu.:19.10          1st Qu.:0.0000
##  Median :  0.3000   Median :24.40          Median :0.0000
##  Mean   :  0.4653   Mean   :24.36          Mean   :0.4323
##  3rd Qu.:  3.7000   3rd Qu.:29.90          3rd Qu.:1.0000
##  Max.   : 12.8000   Max.   :40.00          Max.   :1.0000
##  gp_exacerbations_12m time_to_event_days followup_event
##  Min.   : 0.0         Min.   : 29.0      Min.   :0.0000
##  1st Qu.: 1.0         1st Qu.:215.0      1st Qu.:1.0000
##  Median : 2.0         Median :390.0      Median :1.0000
##  Mean   : 1.9         Mean   :389.7      Mean   :0.9661
##  3rd Qu.: 3.0         3rd Qu.:566.0      3rd Qu.:1.0000
##  Max.   :12.0         Max.   :727.0      Max.   :1.0000
```

The dataset contains 620 observations and 17 variables. The character variables that might be of interest, including *sex*, *smoking_status*, *deprivation_quintile*, and *new_therapy*, would be more convenient to use (and efficient to store) as factors.

The binary variables *severe_exacerbation_12m* and *followup_event* could work as the boolean data type. However, I prefer factors here as well, since factors can be assigned meaningful labels. The common convention for binary data is to code the presence of the outcome event (in this case, severe exacerbation at month 12 and hospital admission/death) as 1 and the absence of such as 0. Both binary variables conform to the convention.

A quick inspection with the *summary()* function shows no issues with missing data.

We can also see considerable variability in age (median = 58, range = 19 to 88) and baseline clinical characteristics of the participants, particularly symptom score (median = 24.9, range = 4.3 to 40.0) and FEV1 (median = 63.1, range = 25.0 to 102.6) Some individuals also had substantially more comorbidities (median = 1, range = 0 to 5) or experienced exacerbations within the previous 12 months (median = 1, range = 0 to 7) much more frequently than others.

```
##    patient_id         age             sex      smoking_status
##  Min.   :  1.0   Min.   :19.00   Female:322   Current:131
##  1st Qu.:155.8   1st Qu.:48.98   Male  :298   Former :246
##  Median :310.5   Median :58.05                Never  :243
##  Mean   :310.5   Mean   :58.29
##  3rd Qu.:465.2   3rd Qu.:67.20
##  Max.   :620.0   Max.   :88.00
##  deprivation_quintile comorbidity_count prior_exacerbations_12m
##  Q1: 88               Min.   :0.000     Min.   :0.000
##  Q2:124               1st Qu.:0.000     1st Qu.:0.000
##  Q3:143               Median :1.000     Median :1.000
```

Age and Baseline Clinical Characteristics by Treatment Group

Treatment Group	Mean Age	SD Age	Mean Symptom Score	SD Symptom Score	Mean FEV1 (%)	SD FEV1 (%)
New therapy	57.1	13.0	25.5	6.59	63.3	11.7
Usual care	59.1	13.1	24.5	6.32	62.5	11.3

Comorbidities and Prior Exacerbations by Treatment Group

Treatment Group	Mean Comorbidities	SD Comorbidities	Mean Prior Exacerbations	SD Prior Exacerbations
New therapy	1.41	1.24	1.49	1.21
Usual care	1.08	1.06	1.10	1.06

```
##  Q4:136               Mean   :1.216     Mean   :1.258
##  Q5:129               3rd Qu.:2.000     3rd Qu.:2.000
##                       Max.   :5.000     Max.   :7.000
##  baseline_symptom_score baseline_fev1_pct      new_therapy  followup_months
##  Min.   : 4.30          Min.   : 25.00    New therapy:249   Min.   : 9.00
##  1st Qu.:20.70          1st Qu.: 55.67    Usual care :371   1st Qu.:10.60
##  Median :24.90          Median : 63.05                      Median :12.20
##  Mean   :24.89          Mean   : 62.80                      Mean   :12.14
##  3rd Qu.:29.23          3rd Qu.: 70.12                      3rd Qu.:13.70
##  Max.   :40.00          Max.   :102.60                      Max.   :15.00
##  symptom_change_6m  followup_symptom_score severe_exacerbation_12m
##  Min.   :-16.3000   Min.   : 2.90          No :352
##  1st Qu.: -2.4000   1st Qu.:19.10          Yes:268
##  Median :  0.3000   Median :24.40
##  Mean   :  0.4653   Mean   :24.36
##  3rd Qu.:  3.7000   3rd Qu.:29.90
##  Max.   : 12.8000   Max.   :40.00
##  gp_exacerbations_12m time_to_event_days followup_event    treatment_group
##  Min.   : 0.0         Min.   : 29.0      No : 21        New therapy:249
##  1st Qu.: 1.0         1st Qu.:215.0      Yes:599        Usual care :371
##  Median : 2.0         Median :390.0
##  Mean   : 1.9         Mean   :389.7
##  3rd Qu.: 3.0         3rd Qu.:566.0
##  Max.   :12.0         Max.   :727.0
```

All of these factors are worth accounting for in later modelling as, for instance, younger age may turn out to be linked to more favorable outcomes and so may better health condition at baseline.

Age, mean symptom score, and FEV1 at baseline seem comparable between groups.

However, the mean comorbidity count and mean exacerbation count within 12 months prior to the study were substantially higher among in the new therapy group.

The differences were formally assessed with the Welch's two-sample t-test. The test produced a p-value < 0.001 for both mean comorbidity count and mean prior exacerbation count, rejecting the null hypotheses of no difference between the group means. These differences are important to adjust for in further modelling, since more favorable health conditions at baseline could be linked to superior treatment outcomes.

	Sex		
	Female	Male	**Total**
Treatment Group			
New therapy	139 (56%)	110 (44%)	249 (100%)
Usual care	183 (49%)	188 (51%)	371 (100%)
Total	322 (52%)	298 (48%)	620 (100%)

	Smoking Status			
	Current	Former	Never	**Total**
Treatment Group				
New therapy	60 (24%)	95 (38%)	94 (38%)	249 (100%)
Usual care	71 (19%)	151 (41%)	149 (40%)	371 (100%)
Total	131 (21%)	246 (40%)	243 (39%)	620 (100%)

```
##
##  Welch Two Sample t-test
##
## data:  comorbidity_count by treatment_group
## t = 3.4421, df = 472.48, p-value = 0.0006285
## alternative hypothesis: true difference in means between group New therapy and group Usual care is no
## 95 percent confidence interval:
##  0.1416537 0.5185396
## sample estimates:
## mean in group New therapy  mean in group Usual care
##                  1.413655                  1.083558
```

```
##
##  Welch Two Sample t-test
##
## data:  prior_exacerbations_12m by treatment_group
## t = 4.1734, df = 482.56, p-value = 3.56e-05
## alternative hypothesis: true difference in means between group New therapy and group Usual care is no
## 95 percent confidence interval:
##  0.2086287 0.5798622
## sample estimates:
## mean in group New therapy  mean in group Usual care
##                  1.493976                  1.099730
```

The new therapy group had a somewhat larger percentage of females than males (56% vs. 44%), while in the standard care group the gender distribution was more even (49% females vs. 51% males). The new therapy group also had a higher percentage of current smokers than the usual care group (24% vs. 19%). Although the Chi-square test of independence yielded no evidence of relationship between group assignment and gender or group assignment and smoking status, both of these variables may still be worth including in the models as potential confounders.

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
```

	Deprivation Quintile					
	Q1	Q2	Q3	Q4	Q5	**Total**
Treatment Group						
New therapy	36 (14%)	48 (19%)	56 (22%)	54 (22%)	55 (22%)	249 (100%)
Usual care	52 (14%)	76 (20%)	87 (23%)	82 (22%)	74 (20%)	371 (100%)
Total	88 (14%)	124 (20%)	143 (23%)	136 (22%)	129 (21%)	620 (100%)

```
## data:  as.data.frame.matrix(table(resp_data$treatment_group, resp_data$sex))
## X-squared = 2.2661, df = 1, p-value = 0.1322
```

```
##
##  Pearson's Chi-squared test
##
## data:  as.data.frame.matrix(table(resp_data$treatment_group, resp_data$smoking_status))
## X-squared = 2.1989, df = 2, p-value = 0.3331
```

The groups were comparable by socio-economic status, with similar distribution of the subject across deprivation quintiles.

Part II

Construct a directed-acyclic-graph and use an appropriate statistical model to investigate whether 'new_therapy' is associated with greater improvement in symptoms at 6 months. You should define your outcome clearly and justify any set of adjustment variables you use. You should also ensure you report and interpret the estimated effect, commenting on any model assumptions and diagnostic checks you consider relevant.

Part III

Is there evidence that 'new_therapy' is associated with fewer GP-treated exacerbations over 12 months? Construct an appropriate statistical model and report an interpretable rate. Ensure you assess whether your chosen model is appropriate, addressing any concerns regarding overdispersion.

Part IV

Explore potential evidence for an association between 'new_therapy' and a longer time to hospital admission or death. In doing so ensure you define clearly the survival outcome and event variables. Comment on censoring and the proportional-hazards assumption when modelling. Additionally, you should consider and discuss whether there are potential interactions between the treatment and any other explanatory variables but are not expected to fit an interaction model.